

RISHAV RAJ

AI Engineer | Agentic AI, RAG & LLM Evaluation

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Summary

M.Sc. Computer Science graduate with hands-on experience building and evaluating agentic AI systems. Project experience with LangGraph agents, RAG pipelines, LLM fine-tuning, and Python applications, with a focus on testing system behaviour under real, adversarial, and edge-case inputs.

Technical Skills

Agentic AI & LLM Systems: LangGraph agents, RAG pipelines, prompt engineering, function/tool calling, structured outputs, LLM fine-tuning (LoRA, GRPO), Hugging Face

Programming & Backend: Python, SQL, FastAPI, pandas, NumPy, Git, GitHub Actions

AI Evaluation & Testing: adversarial and prompt-injection testing, automated test suites, model evaluation, PR-AUC, ROC-AUC

Databases & Vector Stores: MySQL, DBMS, Pinecone, embeddings, document retrieval

Development & Deployment: Docker, AWS Lambda, OOP, automated testing

Selected Projects

Sentinel Procurement | LangGraph, Python

Jun – Aug 2026

github.com/dLounce/sentinel-procurement

- Built an agentic procurement system in LangGraph that separated vendor-text interpretation from purchase approval and checked budget, price, and delivery rules before an order could be created.
- Evaluated the agent under adversarial conditions: ran 100 controlled prompt-injection trials with no unauthorized orders, and compared matching offers with and without injected instructions.
- Wrote 310 tests for offer parsing, purchasing rules, and negotiation behaviour, including malformed inputs and attempts to bypass approval checks.

Text-to-SQL Fine-Tuning | Python, LoRA, GRPO

Mar – Jun 2026

github.com/dLounce/Text-to-SQL | Live demo

- Improved SQL execution accuracy from 50.6% to 68.4% on 1,600 questions across 21 held-out databases by fine-tuning Qwen2.5-Coder-7B with LoRA and GRPO.
- Used execution-based rewards and selected GRPO prompts where the SFT model produced mixed results across eight sampled outputs; kept training and evaluation databases separate.
- Deployed an AWS Lambda demo with query-time limits and checks against reference query results.

Multimodal Deepfake Detection | PyTorch, CLIP

Dec 2025 – Feb 2026

github.com/dLounce/dgm4-multimodal-detection

- Improved image-caption deepfake-detection AUC from 0.863 to 0.965 over a frozen CLIP baseline by training LoRA adapters and cross-attention layers on CLIP ViT-L/14.
- Reached 0.911 mAP for manipulation-type classification and 0.892 mean IoU for face localization using cross-attention features and a classifier over candidate face crops.
- Generated detection explanations with LLaVA-7B using image, face, and caption evidence.

Credit Card Fraud Detection | Python, scikit-learn, XGBoost

Oct – Nov 2025

github.com/dLounce/Fraud-Detection

- Trained an XGBoost classifier on 284,807 anonymized credit-card transactions, where fraudulent transactions made up 0.172% of the dataset.
- Used a 60/20/20 stratified train, validation, and test split; tuned parameters with 5-fold stratified cross-validation and RandomizedSearchCV, using average precision for model selection.
- Chose the decision threshold on validation data and achieved 0.869 PR-AUC and 0.979 ROC-AUC on the held-out test set, with 0.90 precision, 0.83 recall, and 0.86 F1.

Experience

Artificial Intelligence Intern | Ballistic Learning Systems

Jun – Jul 2025

Faridabad, Haryana, India

- Improved agreement between an LLM grader and human graders from 44% to 71% by hand-labeling 118 answers and testing prompt changes against that set.
- Added PDF-based reference material to grading and question generation by chunking documents, storing embeddings in Pinecone, and retrieving relevant content for prompts.
- Designed a MySQL schema for quiz sessions, scores, and grading history.

Education

M.Sc. in Computer Science

Jul 2024 – Jul 2026

Central University of Rajasthan, Ajmer | CGPA: 7.1/10

Bachelor of Computer Applications

Aug 2018 – Jul 2021

Patliputra University, Patna | 73%

Certifications & Achievements: UGC NET Computer Science, Qualified (Aug 2026) | Hugging Face Agents Course (2026)

AnalytiX DataOps Hackathon, CURaj (2025)